JEE MAIN / PREVIOUS YEAR PAPER

JEE Main 2026 22nd January Morning Shift

Question Paper with Answer Key

75	ENGLISH	DARK
QUESTIONS	LANGUAGE	FORMAT

INSIDE THIS PDF

Chemistry	25 questions
Mathematics	25 questions
Physics	25 questions
Answer key	Included at the end

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Chemistry 25 questions

Q1. MCQ (Single Correct Answer)

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Given below are two statements :

Statement I : Sucrose is dextrorotatory. However, sucrose upon hydrolysis gives a solution having mixture of products. This solution shows laevorotation.

Statement II : Hydrolysis of sucrose gives glucose and fructose. Since the laevorotation of glucose is more than the dextrorotation of fructose, the resulting solution becomes laevorotatory.

In the light of the above statements, choose the correct answer from the options given below :

(A) Both Statement I and Statement II are false

(B) Both Statement I and Statement II are true

(C) Statement I is true but Statement II is false

(D) Statement I is false but Statement II is true

Q2. MCQ (Single Correct Answer)

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Given below are two statements :

Statement I : Phenol on treatment with $CHCl_3$ /aq. KOH under refluxing condition, followed by acidification produces p-hydroxy benzaldehyde as the major product and o-hydroxy benzaldehyde as the minor product.

Statement II : The mixture of p-hydroxybenzaldehyde and o-hydroxybenzaldehyde can be easily separated through steam distillation.

In the light of the above statements, choose the correct answer from the options given below

(A) Statement I is true but Statement II is false

(B) Both Statement I and Statement II are true

(C) Both Statement I and Statement II are false

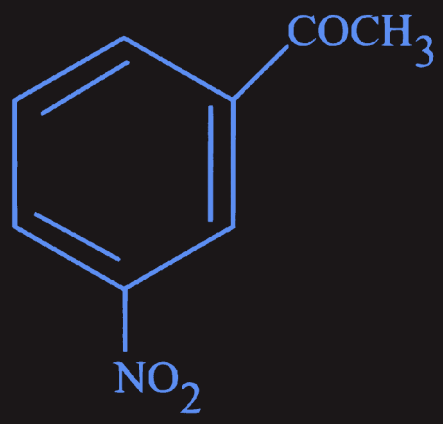
(D) Statement I is false but Statement II is true

Q3. MCQ (Single Correct Answer)

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Given below are two statements :

Statement I : Benzene is nitrated to give nitrobenzene, which on further treatment with $CH_3COCl/AlCl_3$ will give



Statement II : $-NO_2$ group is a m-directing, and deactivating group.

In the light of the above statements, choose the most appropriate answer from the options given below

(A) Statement I is incorrect but Statement II is correct

(B) Both Statement I and Statement II are correct

(C) Both Statement I and Statement II are incorrect

(D) Statement I is correct but Statement II is incorrect

Q4. MCQ (Single Correct Answer)

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In the reaction,

$2Al(s) + 6HCl(aq) \rightarrow 2Al^{3+}(aq) + 6Cl^{-}(aq) + 3H_2(g)$

(A) $12\,L\,HCl(aq)$ is consumed for every $6\,L\,H_2(g)$ produced.

(B) $11.2\,L\,H_2(g)$ at STP is produced for every mole of HCl consumed.

(C) $33.6\,L\,H_2(g)$ is produced regardless of temperature and pressure for every mole of Al that reacts.

(D) $67.2\,L\,H_2(g)$ at STP is produced for every mole of Al that reacts.

Q5. MCQ (Single Correct Answer)

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As compared with chlorocyclohexane, which of the following statements correctly apply to chlorobenzene?

A. The magnitude of negative charge is more on chlorine atom.

B. The $\mathrm{C}-\mathrm{Cl}$ bond has partial double bond character.

C. $\mathrm{C}-\mathrm{Cl}$ bond is less polar.

D. $\mathrm{C}-\mathrm{Cl}$ bond is longer due to repulsion between delocalised electrons of the aromatic ring and lone pairs of electrons of chlorine.

E. The $\mathrm{C}-\mathrm{Cl}$ bond is formed using sp^2 hybridised orbital of carbon.

Choose the correct answer from the options given below :

(A) B, C and E Only (B) A, C and E Only (C) B, C and D Only (D) A, D and E Only

Q6. MCQ (Single Correct Answer)

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Given below are two statements :

Statement I : The halogen that makes longest bond with hydrogen in HX, has the smallest covalent radius in its group.

Statement II : A group 15 element's hydride EH_3 has the lowest boiling point among corresponding hydrides of other group 15 elements. The maximum covalency of that element E is 4 .

In the light of the above statements, choose the correct answer from the options given below

(A) Statement I is true but Statement II is false

(B) Statement I is false but Statement II is true

(C) Both Statement I and Statement II are true

(D) Both Statement I and Statement II are false

Q7. MCQ (Single Correct Answer)

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The energy required by electrons, present in the first Bohr orbit of hydrogen atom to be excited to second Bohr orbit is ____ Jmol^{-1}.

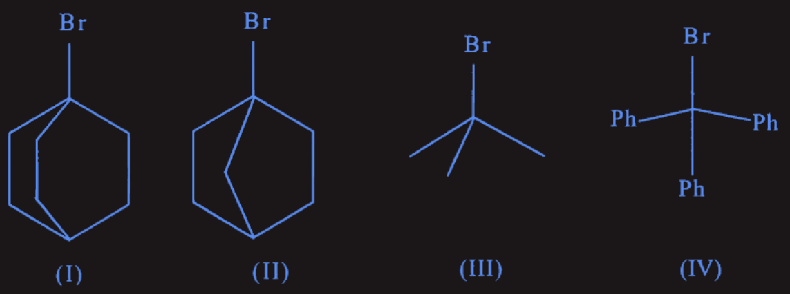
Given: $R_H = 2.18 \times 10^{-11}\mathrm{ergs}$.

(A) 9.835×10^{12} (B) 1.635×10^{-11} (C) 1.635×10^{-18} (D) 9.835×10^{5}

Q8. MCQ (Single Correct Answer)

The correct order of the rate of reaction of the following reactants with nucleophile by S_N1 mechanism is :

(Given : Structures I and II are rigid)



(A) I $<$ II $<$ III $<$ IV

(B) II $<$ I $<$ III $<$ IV

(C) III $<$ I $<$ II $<$ IV

(D) IV $<$ III $<$ II $<$ I

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Q9. MCQ (Single Correct Answer)

Consider the transition metal ions Mn^{3+}, Cr^{3+}, Fe^{3+} and Co^{3+} and all form low spin octahedral complexes. The correct decreasing order of unpaired electrons in their respective d-orbitals of the complexes is

(A) $Cr^{3+} > Mn^{3+} > Fe^{3+} > Co^{3+}$

(B) $Mn^{3+} > Fe^{3+} > Co^{3+} > Cr^{3+}$

(C) $Fe^{3+} > Co^{3+} > Mn^{3+} > Cr^{3+}$

(D) $Cr^{3+} > Fe^{3+} > Co^{3+} > Mn^{3+}$

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Q10. MCQ (Single Correct Answer)

$A \rightarrow$ product (First order reaction).

Three sets of experiment were performed for a reaction under similar experimental conditions:

Run $1 \Rightarrow 100\,mL$ of 10 M solution of reactant A

Run $2 \Rightarrow 200\,mL$ of 10 M solution of reactant A

Run $3 \Rightarrow 100\,mL$ of 10 M solution of reactant $A + 100\,mL$ of H_2O added.

The correct variation of rate of reaction is

(A) Run 1 = Run 2 = Run 3

(B) Run $3 <$ Run $1 <$ Run 2

(C) Run $1 <$ Run $2 <$ Run 3

(D) Run $3 <$ Run $1 =$ Run 2

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Q11. MCQ (Single Correct Answer)

A 'p'-block element (E) and hydrogen form a binary cation $(EH_x)^+$, while EH_3 on treatment with K_2HgI_4 in alkaline medium gives a precipitate of basic mercury(II)amido-iodine. Given below are first ionisation enthalpy values $(kJmol^{-1})$ for first element each from group 13, 14, 15 and 16. Identify the correct first ionisation enthalpy value for element E.

(A) 1402 (B) 1086 (C) 801 (D) 1312

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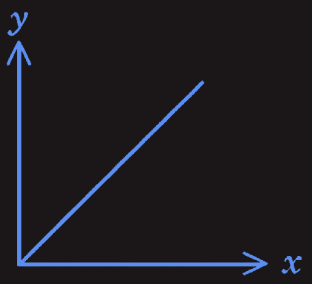
Q12. MCQ (Single Correct Answer)

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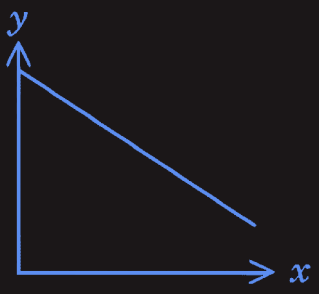
Consider a solution of $CO_2(g)$ dissolved in water in a closed container.

Which one of the following plots correctly represents variation of log (partial pressure of CO_2 in vapour phase above water) [y-axis] with log (mole fraction of CO_2 in water) [x-axis] at $25^\circ C$?

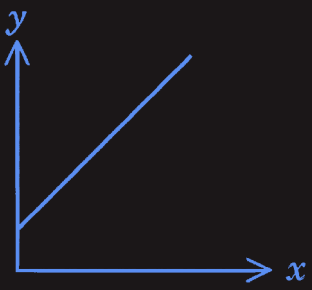
(A)



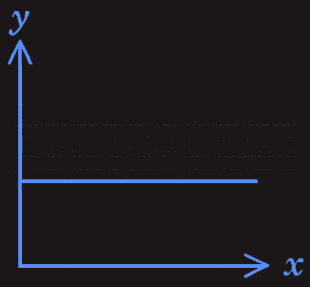
(B)



(C)



(D)



Q13. MCQ (Single Correct Answer)

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Match the LIST-I with LIST-II

List-I Reagents		List-II Name of Reaction involving carbonyl compounds	
A.	$NH_2 - NH_2,\ KOH$	I.	Tollen's Test
B.	$Ag(NH_3)_2OH$	II.	Clemmensen Reduction
C.	Aq. $CuSO_4$, Sodium Potassium tartrate, KOH	III.	Wolff - Kishner Reduction
D.	$Zn - Hg,\ HCl$	IV.	Fehling's Test

Choose the correct answer from the options given below :

(A) A-IV, B-III, C-II, D-I

(B) A-III, B-I, C-IV, D-II

(C) A-II, B-I, C-IV, D-III

(D) A-III, B-IV, C-I, D-II

Q14. MCQ (Single Correct Answer)

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Given below are two statements :

Statement I : The Henry's law constant $\mathrm{K_H}$ is constant with respect to variations in solution's concentration over the range for which the solution is ideally dilute.

Statement II : $\mathrm{K_H}$ does not differ for the same solute in different solvents.

In the light of the above statements, choose the correct answer from the options given below

(A) Both Statement I and Statement II are true

(B) Statement I is true but Statement II is false

(C) Statement I is false but Statement II is true

(D) Both Statement I and Statement II are false

Q15. MCQ (Single Correct Answer)

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The correct order of reactivity of $\mathrm{CH_3Br}$ in methanol with the following nucleophiles is

$\mathrm{F^-, I^-, C_2H_5O^-}$ and $\mathrm{C_6H_5O^-}$

(A) $\mathrm{I^- > C_2H_5O^- > C_6H_5O^- > F^-}$

(B) $\mathrm{I^- > F^- > C_6H_5O^- > C_2H_5O^-}$

(C) $\mathrm{I^- > C_6H_5O^- > F^- > C_2H_5O^-}$

(D) $\mathrm{I^- > C_2H_5O^- > F^- > C_6H_5O^-}$

Q16. MCQ (Single Correct Answer)

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Two p-block elements X and Y form fluorides of the type EF_3. The fluoride compound XF_3 is a Lewis acid and YF_3 is a Lewis base. The hybridizations of the central atoms of XF_3 and YF_3 respectively are

(A) sp^2 and sp^3

(B) Both sp^3

(C) Both sp^2

(D) sp^3 and sp^2

Q17. MCQ (Single Correct Answer)

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A first row transition metal (M) does not liberate H_2 gas from dilute HCl .1 mol of aqueous solution of MSO_4 is treated with excess of aqueous KCN and then $H_2S(g)$ is passed through the solution. The amount of MS (metal sulphide) formed from the above reaction is ____ mol.

(A) 1 (B) 2 (C) 0 (D) 3

Q18. MCQ (Single Correct Answer)

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Match the LIST-I with LIST-II

List-I **Thermodynamic Process**	**List-II** **Magnitude in kJ**
A. Work done in reversible, isothermal expansion of 2 mol of ideal gas from $2\,dm^3$ to $20\,dm^3$ at $300\,K$.	I. 4
B. Work done in irreversible isothermal expansion of 1 mol ideal gas from $1\,m^3$ to $3\,m^3$ at $300\,K$ against a constant pressure of $3\,kPa$.	II. 11.5
C. Change in internal energy for adiabatic expansion of a 1 mol ideal gas with change of temperature $= 320\,K$ and $\overline{C}_V = \frac{3}{2}R$.	III. 6
D. Change in enthalpy at constant pressure of 1 mol ideal gas with change of temperature $= 337\,K$ and $\overline{C}_p = \frac{5}{2}R$.	IV. 7

(A) A-I, B-II, C-III, D-IV

(B) A-III, B-II, C-IV, D-I

(C) A-II, B-I, C-III, D-IV

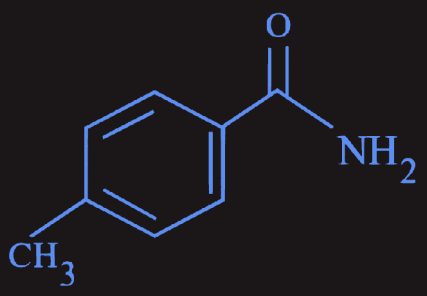
(D) A-II, B-III, C-I, D-IV

Q19. MCQ (Single Correct Answer)

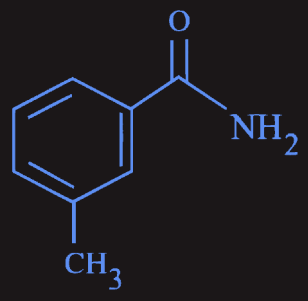
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'A' is a neutral organic compound (M. F : C_8H_9ON). On treatment with aqueous $Br_2/HO^{(-)}$, ' A ' forms a compound ' B ' which is soluble in dilute acid. ' B ' on treatment with aqueous $NaNO_2/HCl\,(0-5^\circ C)$ produces a compound ' C ' which on treatment with $CuCN/NaCN$ produces ' D '. Hydrolysis of ' D ' produces ' E ' which is also obtainable from the hydrolysis of ' A '. ' E ' on treatment with acidified $KMnO_4$ produces ' F '. ' F ' contains two different types of hydrogen atoms. The structure of ' A ' is

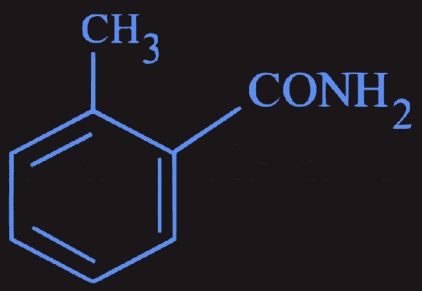
(A)



(B)



(C)



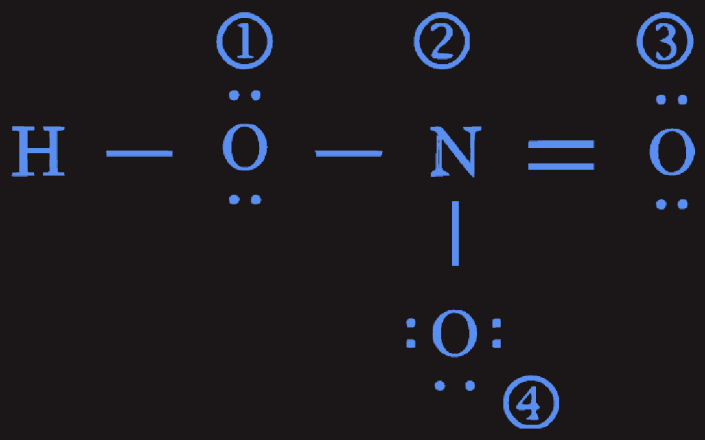
(D)



Q20. MCQ (Single Correct Answer)

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The formal charges on the atoms marked as (1) to (4) in the Lewis representation of $\mathrm{HNO_3}$ molecule respectively are



(A) $0, 0, -1, +1$ (B) $0, -1, 0, +1$ (C) $0, +1, 0, -1$ (D) $+1, 0, 0, -1$

Q21. Numerical

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Sodium fusion extract of an organic compound (Y) with $\mathrm{CHCl_3}$ and chlorine water gives violet color to the $\mathrm{CHCl_3}$ layer. 0.15 g of (Y) gave 0.12 g of the silver halide precipitate in Carius method. Percentage of halogen in the compound (Y) is

____ . (Nearest integer)

(Given : molar mass $\mathrm{gmol^{-1}} \mathrm{C} : 12, \mathrm{H} : 1, \mathrm{Cl} : 35.5, \mathrm{Br} : 80, \mathrm{I} : 127$)

Q22. Numerical

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The cycloalkene (X) on bromination consumes one mole of bromine per mole of (X) and gives the product (Y) in which $\mathrm{C} : \mathrm{Br}$ ratio is 3:1. The percentage of bromine in the product (Y) is ____ %. (Nearest integer)

(Given : molar mass in $\mathrm{gmol^{-1}} \mathrm{H} : 1, \mathrm{C} : 12, \mathrm{O} : 16, \mathrm{Br} : 80$)

Q23. Numerical

Dissociation of a gas A_2 takes place according to the following chemical reaction. At equilibrium, the total pressure is 1 bar at 300 K .

$A_2(g) \rightleftharpoons 2\,A(g)$

The standard Gibbs energy of formation of the involved substances has been provided below:

Substance	$\Delta G_f^\circ / \mathrm{kJ\ mol^{-1}}$
A_2	-100.00
A	-50.832

The degree of dissociation of $A_2(g)$ is given by $\left(x \times 10^{-2}\right)^{1/2}$ where $x =$ ____ . (Nearest integer).

[**Given :** $R = 8\ \mathrm{J\ mol^{-1}\ K^{-1}}, \log 2 = 0.3010, \log 3 = 0.48$]

Assume degree of dissociation is not negligible.

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Q24. Numerical

The temperature at which the rate constants of the given below two gaseous reactions become equal is ____ K. (Nearest integer)

$X \longrightarrow Y, \quad k_1 = 10^6 e^{\frac{-30000}{T}}$

$P \longrightarrow Q, \quad k_2 = 10^4 e^{\frac{-24000}{T}}$

Given : $\ln 10 = 2.303$

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Q25. Numerical

Consider the following electrochemical cell at 298 K

$Pt\,|HSnO_2^-(aq)|Sn(OH)_6{}^{2-}(aq)\,|OH^-(aq)|Bi_2O_3(s)\,|\,Bi(s).$

If the reaction quotient at a given time is 10^6, then the cell EMF $\left(E_{cell}\right)$ is ____ $\times 10^{-1}$ V (Nearest integer).

Given the standard half-cell reduction potential as

$E^\circ_{Bi_2O_3/Bi,OH^-} = -0.44$ V and $E^\circ_{Sn(OH)_6^{2-}/HSnO_2^-,OH^-} = -0.90$ V

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Mathematics 25 questions

Q26. MCQ (Single Correct Answer)

Two distinct numbers a and b are selected at random from $1, 2, 3, \ldots, 50$. The probability, that their product ab is divisible by 3 , is

(A) $\frac{272}{1225}$ (B) $\frac{561}{1225}$ (C) $\frac{664}{1225}$ (D) $\frac{8}{25}$

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Q27. MCQ (Single Correct Answer)

The number of solutions of $\tan^{-1} 4x + \tan^{-1} 6x = \frac{\pi}{6}$, where $-\frac{1}{2\sqrt{6}} < x < \frac{1}{2\sqrt{6}}$, is equal to :

(A) 2 (B) 0 (C) 3 (D) 1

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Q28. MCQ (Single Correct Answer)

If the line $\alpha x + 2y = 1$, where $\alpha \in \mathbb{R}$, does not meet the hyperbola $x^2 - 9y^2 = 9$, then a possible value of α is :

(A) 0.6 (B) 0.7 (C) 0.8 (D) 0.5

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Q29. MCQ (Single Correct Answer)

The number of distinct real solutions of the equation $x|x + 4| + 3|x + 2| + 10 = 0$ is

(A) 2 (B) 3 (C) 0 (D) 1

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Q30. MCQ (Single Correct Answer)

Let the relation R on the set $\mathrm{M} = \{1, 2, 3, \ldots, 16\}$ be given by $\mathrm{R} = \{(x, y) : 4y = 5x - 3, x, y \in \mathrm{M}\}$.

Then the minimum number of elements required to be added in R , in order to make the relation symmetric, is equal to

(A) 4 (B) 3 (C) 1 (D) 2

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Q31. MCQ (Single Correct Answer)

Let the set of all values of r, for which the circles $(x + 1)^2 + (y + 4)^2 = r^2$ and $x^2 + y^2 - 4x - 2y - 4 = 0$ intersect at two distinct points be the interval (α, β). Then $\alpha\beta$ is equal to

(A) 21 (B) 24 (C) 20 (D) 25

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Q32. MCQ (Single Correct Answer)

Let the solution curve of the differential equation $xdy - ydx = \sqrt{x^2 + y^2}dx, x > 0, y(1) = 0$, be $y = y(x)$. Then $y(3)$ is equal to

(A) 6 (B) 4 (C) 1 (D) 2

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Q33. MCQ (Single Correct Answer)

If the chord joining the points $\mathrm{P}_1(x_1, y_1)$ and $\mathrm{P}_2(x_2, y_2)$ on the parabola $y^2 = 12x$ subtends a right angle at the vertex of the parabola, then $x_1x_2 - y_1y_2$ is equal to

(A) 280 (B) 288 (C) 292 (D) 284

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Q34. MCQ (Single Correct Answer)

If $A=\begin{bmatrix}2 & 3\\ 3 & 5\end{bmatrix}$, then the determinant of the matrix $\left(A^{2025}-3A^{2024}+A^{2023}\right)$ is

(A) 12 (B) 24 (C) 28 (D) 16

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Q35. MCQ (Single Correct Answer)

The value of $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\left(\frac{1}{[x]+4}\right)dx$, where $[\cdot]$ denotes the greatest integer function, is

(A) $\frac{1}{60}(21\pi-1)$ (B) $\frac{1}{60}(\pi-7)$ (C) $\frac{7}{60}(\pi-3)$ (D) $\frac{7}{60}(3\pi-1)$

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Q36. MCQ (Single Correct Answer)

Let $\overrightarrow{AB}=2\hat{i}+4\hat{j}-5\hat{k}$ and $\overrightarrow{AD}=\hat{i}+2\hat{j}+\lambda\hat{k}, \lambda\in\mathbb{R}$. Let the projection of the vector $\vec{v}=\hat{i}+\hat{j}+\hat{k}$ on the diagonal $\overrightarrow{AC}$ of the parallelogram ABCD be of length one unit. If α, β, where $\alpha>\beta$, be the roots of the equation $\lambda^2x^2-6\lambda x+5=0$, then $2\alpha-\beta$ is equal to

(A) 3 (B) 6 (C) 4 (D) 1

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Q37. MCQ (Single Correct Answer)

Let the line $x=-1$ divide the area of the region $\{(x,y): 1+x^2\le y\le 3-x\}$ in the ratio $m:n, \gcd(m,n)=1$. Then $m+n$ is equal to

(A) 27 (B) 28 (C) 25 (D) 26

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Q38. MCQ (Single Correct Answer)

Let $f(x)=x^{2025}-x^{2000}, x\in[0,1]$ and the minimum value of the function $f(x)$ in the interval $[0,1]$ be $(80)^{80}(n)^{-81}$. Then n is equal to

(A) -40 (B) -41 (C) -80 (D) -81

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Q39. MCQ (Single Correct Answer)

If the sum of the first four terms of an A.P. is 6 and the sum of its first six terms is 4 , then the sum of its first twelve terms is

(A) -26 (B) -20 (C) -24 (D) -22

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Q40. MCQ (Single Correct Answer)

If a random variable x has the probability distribution

x	0	1	2	3	4	5	6	7
$\mathrm{P}(x)$	0	2k	k	3k	$2\mathrm{k}^2$	2k	$\mathrm{k}^2+\mathrm{k}$	$7\mathrm{k}^2$

then $\mathrm{P}(3 < x \leq 6)$ is equal to

(A) 0.34 (B) 0.64 (C) 0.22 (D) 0.33

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Q41. MCQ (Single Correct Answer)

The coefficient of x^{48} in $(1+x)+2(1+x)^2+3(1+x)^3+\ldots+100(1+x)^{100}$ is equal to

(A) $100 \cdot {}^{100}\mathrm{C}_{49} - {}^{100}\mathrm{C}_{48}$ (B) $100 \cdot {}^{101}\mathrm{C}_{49} - {}^{101}\mathrm{C}_{50}$

(C) ${}^{100}\mathrm{C}_{50} + {}^{101}\mathrm{C}_{49}$ (D) $100 \cdot {}^{100}\mathrm{C}_{49} - {}^{100}\mathrm{C}_{50}$

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Q42. MCQ (Single Correct Answer)

Let $\mathrm{P}(\alpha,\beta,\gamma)$ be the point on the line $\frac{x-1}{2}=\frac{y+1}{-3}=z$ at a distance $4\sqrt{14}$ from the point $(1,-1,0)$ and nearer to the origin. Then the shortest distance, between the lines $\frac{x-\alpha}{1}=\frac{y-\beta}{2}=\frac{z-\gamma}{3}$ and $\frac{x+5}{2}=\frac{y-10}{1}=\frac{z-3}{1}$, is equal to

(A) $4\sqrt{\frac{7}{5}}$ (B) $7\sqrt{\frac{5}{4}}$ (C) $4\sqrt{\frac{5}{7}}$ (D) $2\sqrt{\frac{7}{4}}$

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Q43. MCQ (Single Correct Answer)

If the domain of the function $f(x)=\sin^{-1}\left(\frac{5-x}{3+2x}\right)+\frac{1}{\log_e(10-x)}$ is $(-\infty,\alpha]\cup[\beta,\gamma)-\{\delta\}$, then $6(\alpha+\beta+\gamma+\delta)$ is equal to

(A) 66 (B) 68 (C) 70 (D) 67

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Q44. MCQ (Single Correct Answer)

Let $f:[1,\infty)\to\mathbb{R}$ be a differentiable function. If $6\int_1^x f(t)dt = 3xf(x)+x^3-4$ for all $x\geq 1$, then the value of $f(2)-f(3)$ is :

(A) 4 (B) 3 (C) -4 (D) -3

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Q45. MCQ (Single Correct Answer)

If the image of the point $\mathrm{P}(1,2,a)$ in the line $\frac{x-6}{3}=\frac{y-7}{2}=\frac{7-\mathrm{z}}{2}$ is $\mathrm{Q}(5,b,\mathrm{c})$, then $a^2+b^2+c^2$ is equal to

(A) 298 (B) 264 (C) 293 (D) 283

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Q46. Numerical

If $\frac{\cos^2 48^\circ - \sin^2 12^\circ}{\sin^2 24^\circ - \sin^2 6^\circ} = \frac{\alpha + \beta\sqrt{5}}{2}$, where $\alpha, \beta \in \mathbb{N}$, then $\alpha + \beta$ is equal to ____

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Q47. Numerical

Let A be a 3×3 matrix such that $\mathrm{A} + \mathrm{A}^{\mathrm{T}} = \mathrm{O}$. If $\mathrm{A}\begin{bmatrix}1\\-1\\0\end{bmatrix} = \begin{bmatrix}3\\3\\2\end{bmatrix}, \mathrm{A}^2\begin{bmatrix}1\\-1\\0\end{bmatrix} = \begin{bmatrix}-3\\19\\-24\end{bmatrix}$ and $\det(\operatorname{adj}(2\operatorname{adj}(\mathrm{A} + \mathrm{I}))) = (2)^\alpha \cdot (3)^\beta \cdot (11)^\gamma, \alpha, \beta, \gamma$ are non-negative integers, then $\alpha + \beta + \gamma$ is equal to ____

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Q48. Numerical

Let ABC be a triangle. Consider four points $\mathrm{p}_1, \mathrm{p}_2, \mathrm{p}_3, \mathrm{p}_4$ on the side AB , five points p_5, p_6, p_7, p_8, p_9 on the side BC, and four points $p_{10}, p_{11}, p_{12}, p_{13}$ on the side AC . None of these points is a vertex of the triangle ABC . Then the total number of pentagons, that can be formed by taking all the vertices from the points $p_1, p_2, \ldots, p_{13}$, is ____

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Q49. Numerical

Let $\alpha = \frac{-1+i\sqrt{3}}{2}$ and $\beta = \frac{-1-i\sqrt{3}}{2}, i = \sqrt{-1}$. If

$(7 - 7\alpha + 9\beta)^{20} + (9 + 7\alpha - 7\beta)^{20} + (-7 + 9\alpha + 7\beta)^{20} + (14 + 7\alpha + 7\beta)^{20} = m^{10}$,

then m is ____

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Q50. Numerical

If $\int (\sin x)^{\frac{-11}{2}} (\cos x)^{\frac{-5}{2}} dx = -\frac{p_1}{q_1}(\cot x)^{\frac{9}{2}} - \frac{p_2}{q_2}(\cot x)^{\frac{5}{2}} - \frac{p_3}{q_3}(\cot x)^{\frac{1}{2}} + \frac{p_4}{q_4}(\cot x)^{\frac{-3}{2}} + \mathrm{C}$, where p_i and q_i are positive integers with $\gcd(p_i, q_i) = 1$ for $i = 1, 2, 3, 4$ and C is the constant of integration, then $\frac{15 p_1 p_2 p_3 p_4}{q_1 q_2 q_3 q_4}$ is equal to ____

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Physics 25 questions

Q51. MCQ (Single Correct Answer)

A thin convex lens of focal length 5 cm and a thin concave lens of focal length 4 cm are combined together (without any gap) and this combination has magnification m_1 when an object is placed 10 cm before the convex lens. Keeping the positions of convex lens and object undisturbed a gap of 1 cm is introduced between the lenses by moving the concave lens away, which lead to a change in magnification of total lens system to m_2. The value of $\left|\frac{m_1}{m_2}\right|$ is ____ .

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(A) $\frac{25}{27}$ (B) $\frac{5}{6}$ (C) $\frac{5}{27}$ (D) $\frac{3}{2}$

Q52. MCQ (Single Correct Answer)

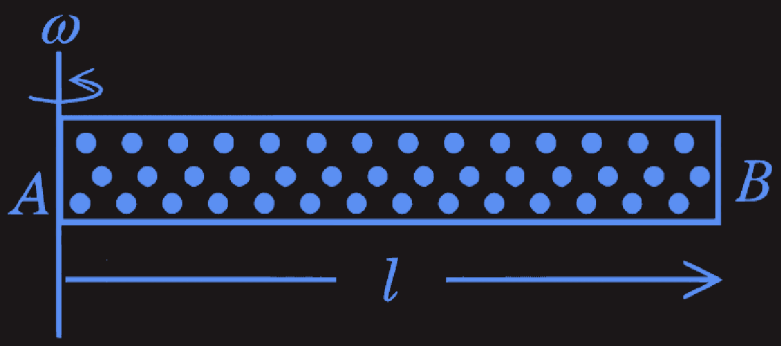
Consider an equilateral prism (refractive index $\sqrt{2}$). A ray of light is incident on its one surface at a certain angle i. If the emergent ray is found to graze along the other surface then the angle of refraction at the incident surface is close to ____

(A) 30° (B) 20° (C) 40° (D) 15°

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Q53. MCQ (Single Correct Answer)

A cylindrical tube AB of length l, closed at both ends contains an ideal gas of 1 mol having molecular weight M. The tube is rotated in a horizontal plane with constant angular velocity ω about an axis perpendicular to AB and passing through the edge at end A, as shown in the figure. If P_A and P_B are the pressures at A and B respectively, then (Consider the temperature is same at all points in the tube)



(A) $P_B = P_A e^{\left(\frac{M\omega^2 l^2}{3RT}\right)}$ (B) $P_B = P_A$

(C) $P_B = P_A e^{\left(\frac{M\omega^2 l^2}{2RT}\right)}$ (D) $P_B = P_A e^{\left(\frac{M\omega^2 l^2}{RT}\right)}$

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Q54. MCQ (Single Correct Answer)

Electric field in a region is given by $\vec{E} = Ax\hat{i} + By\hat{j}$, where $A = 10\ \mathrm{V/m^2}$ and $B = 5\ \mathrm{V/m^2}$. If the electric potential at a point $(10, 20)$ is 500 V , then the electric potential at origin is ____ V.

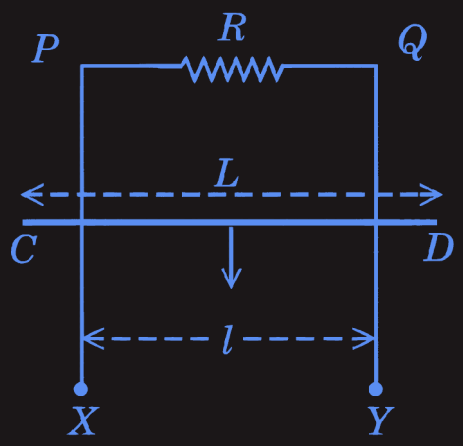
(A) 1000 (B) 0 (C) 2000 (D) 500

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Q55. MCQ (Single Correct Answer)

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$XPQY$ is a vertical smooth long loop having a total resistance R where PX is parallel to QY and separation between them is l. A constant magnetic field B perpendicular to the plane of the loop exists in the entire space. A rod CD of length $L(L > l)$ and mass m is made to slide down from rest under the gravity as shown in figure. The terminal speed acquired by the rod is ____ $\mathrm{m/s}$. (g = acceleration due to gravity)

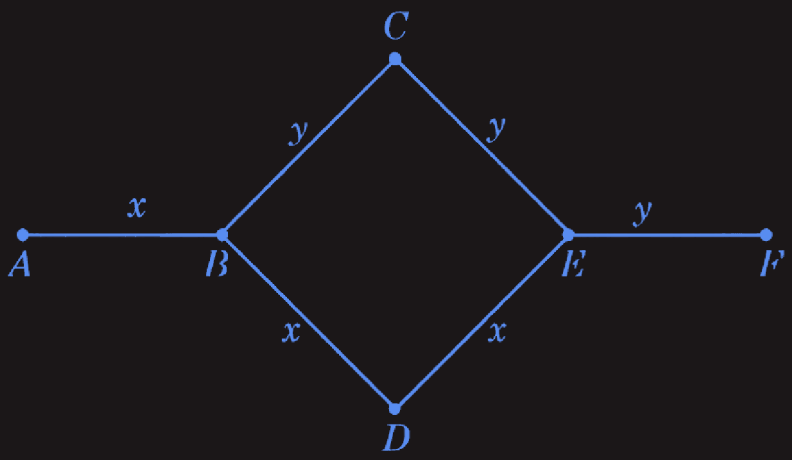


(A) $\frac{m\,gR}{B^2l^2}$ (B) $\frac{2mgR}{B^2L^2}$ (C) $\frac{2mgR}{B^2l^2}$ (D) $\frac{8mgR}{B^2l^2}$

Q56. MCQ (Single Correct Answer)

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Rods x and y of equal dimensions but of different materials are joined as shown in figure. Temperatures of end points A and F are maintained at $100^\circ\mathrm{C}$ and $40^\circ\mathrm{C}$ respectively. Given the thermal conductivity of $\mathrm{rod}\,x$ is three times of that of $\mathrm{rod}\,y$, the temperature at junction points B and E are (close to):



(A) $60^\circ\mathrm{C}$ and $45^\circ\mathrm{C}$ respectively

(B) $80^\circ\mathrm{C}$ and $70^\circ\mathrm{C}$ respectively

(C) $89^\circ\mathrm{C}$ and $73^\circ\mathrm{C}$ respectively

(D) $80^\circ\mathrm{C}$ and $60^\circ\mathrm{C}$ respectively

Q57. MCQ (Single Correct Answer)

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Match the LIST-I with LIST-II

List-I		List-II	
A.	Spring constant	I.	$ML^2T^{-2}K^{-1}$
B.	Thermal conductivity	II.	ML^0T^{-2}
C.	Boltzmann constant	III.	$ML^2T^{-3}A^{-2}$
D.	Inductive reactance	IV.	$MLT^{-3}K^{-1}$

Choose the correct answer from the options given below:

(A) A-III, B-II, C-IV, D-I

(B) A-I, B-IV, C-II, D-III

(C) A-II, B-I, C-IV, D-III

(D) A-II, B-IV, C-I, D-III

Q58. MCQ (Single Correct Answer)

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Given below are two statements :

Statement I : Pressure of a fluid is exerted only on a solid surface in contact as the fluid-pressure does not exist everywhere in a still fluid.

Statement II : Excess potential energy of the molecules on the surface of a liquid, when compared to interior, results in surface tension.

In the light of the above statements, choose the correct answer from the options given below

(A) Statement I is false but Statement II is true

(B) Statement I is true but Statement II is false

(C) Both Statement I and Statement II are false

(D) Both Statement I and Statement II are true

Q59. MCQ (Single Correct Answer)

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A solid sphere of mass 5 kg and radius 10 cm is kept in contact with another solid sphere of mass 10 kg and radius 20 cm . The moment of inertia of this pair of spheres about the tangent passing through the point of contact is ____ $\mathrm{kg \cdot m^2}$.

(A) 0.18

(B) 0.72

(C) 0.36

(D) 0.63

Q60. MCQ (Single Correct Answer)

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A simple pendulum has a bob with mass m and charge q. The pendulum string has negligible mass. When a uniform and horizontal electric field $\vec{E}$ is applied, the tension in the string changes. The final tension in the string, when pendulum attains an equilibrium position is ____ .

(A) $mg - qE$

(B) $\sqrt{m^2g^2 + q^2E^2}$

(C) $m\,\mathrm{g} + qE$

(D) $\sqrt{m^2g^2 - q^2E^2}$

Q61. MCQ (Single Correct Answer)

A projectile is thrown upward at an angle 60° with the horizontal. The speed of the projectile is $20\ \mathrm{m/s}$ when its direction of motion is 45° with the horizontal. The initial speed of the projectile is ____ $\mathrm{m/s}$.

(A) $20\sqrt{3}$ (B) $20\sqrt{2}$ (C) 40 (D) $40\sqrt{2}$

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Q62. MCQ (Single Correct Answer)

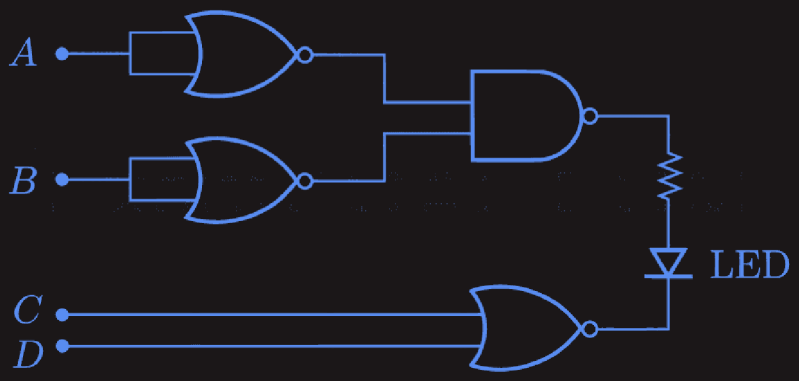
The escape velocity from a spherical planet A is $10\ \mathrm{km/s}$. The escape velocity from another planet B whose density and radius are 10% of those of planet A, is ____ $\mathrm{m/s}$.

(A) 1000 (B) $200\sqrt{5}$ (C) $1000\sqrt{2}$ (D) $100\sqrt{10}$

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Q63. MCQ (Single Correct Answer)

Find the correct combination of $\mathrm{A}, \mathrm{B}, \mathrm{C}$ and D inputs which can cause the LED to glow.



(A) 1000 (B) 0011 (C) 0100 (D) 1101

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Q64. MCQ (Single Correct Answer)

Three identical coils C_1, C_2 and C_3 are closely placed such that they share a common axis. C_2 is exactly midway. C_1 carries current I in anti-clockwise direction while C_3 carries current I in clockwise direction. An induced current flows through C_2 will be in clockwise direction when

(A) C_1 and C_3 move with equal speeds away from C_2

(B) C_1 moves towards C_2 and C_3 moves away from C_2

(C) C_1 and C_3 move with equal speeds towards C_2

(D) C_1 moves away from C_2 and C_3 moves towards C_2

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Q65. MCQ (Single Correct Answer)

The volume of an ideal gas increases 8 times and temperature becomes $(1/4)^{\text{th}}$ of initial temperature during a reversible change. If there is no exchange of heat in this process $(\Delta Q = 0)$ then identify the gas from the following options (Assuming the gases given in the options are ideal gases) :

(A) $\mathrm{NH_3}$ (B) $\mathrm{O_2}$ (C) $\mathrm{CO_2}$ (D) He

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Q66. MCQ (Single Correct Answer)

The minimum frequency of photon required to break a particle of mass 15.348 amu into 4α particles is ____ kHz .

[mass of He nucleus = $4.002\text{amu}, 1\text{amu} = 1.66 \times 10^{-27}\ \text{kg}$, $\text{h} = 6.6 \times 10^{-34}\ \text{J.s}$ and $\text{c} = 3 \times 10^{8}\ \text{m/s}$]

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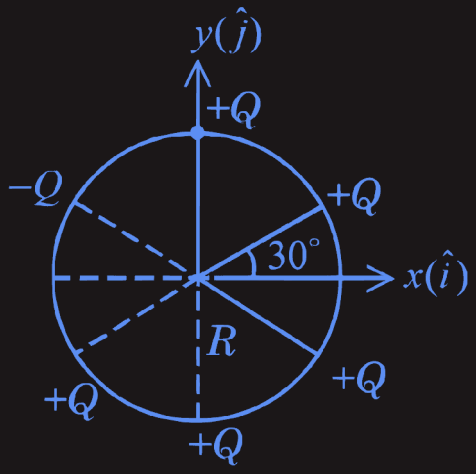
(A) 14.94×10^{20} (B) 9×10^{19} (C) 9×10^{20} (D) 14.94×10^{19}

Q67. MCQ (Single Correct Answer)

Six point charges are kept 60° apart from each other on the circumference of a circle of radius R as shown in figure. The net electric field at the center of the circle is ____ .

(ϵ_0 is permittivity of free space)

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(A) $-\left(\frac{5Q}{8\pi\epsilon_0 R^2}\right)(\hat{i} - 3\hat{j})$ (B) $\frac{Q}{4\pi\epsilon_0 R^2}(\sqrt{3}\hat{i} - \hat{j})$

(C) $-\frac{Q}{4\pi\epsilon_0 R^2}(\sqrt{3}\hat{i} - \hat{j})$ (D) $-\frac{5Q}{8\pi\epsilon_0 R^2}(\hat{i} + \sqrt{3}\hat{j})$

Q68. MCQ (Single Correct Answer)

$7.9\text{MeV}\ \alpha$-particle scatters from a target material of atomic number 79 . From the given data the estimated diameter of nuclei of the target material is (approximately) ____ m.

$\left[\frac{1}{4\pi\epsilon_o} = 9 \times 10^9 \text{Nm}^2/\text{C}^2 \text{ and electron charge } = 1.6 \times 10^{-19}\text{C}\right]$

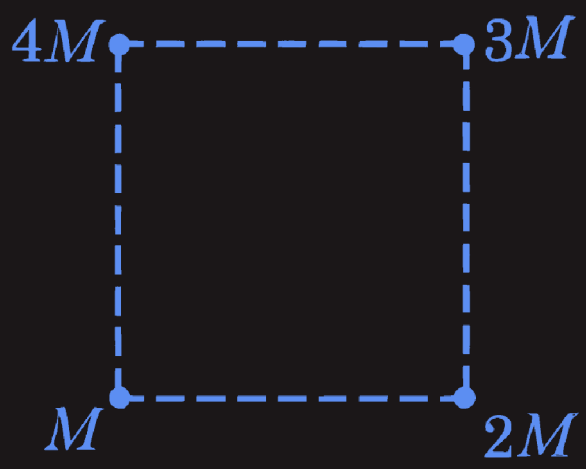
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(A) 2.88×10^{-14} (B) 5.76×10^{-14} (C) 1.44×10^{-13} (D) 1.69×10^{-12}

Q69. MCQ (Single Correct Answer)

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Net gravitational force at the center of a square is found to be F_1 when four particles having mass $M, 2M, 3M$ and $4M$ are placed at the four corners of the square as shown in figure and it is F_2 when the positions of $3M$ and $4M$ are interchanged. The ratio $\frac{F_1}{F_2}$ is $\frac{\alpha}{\sqrt{5}}$. The value of α is ____ .

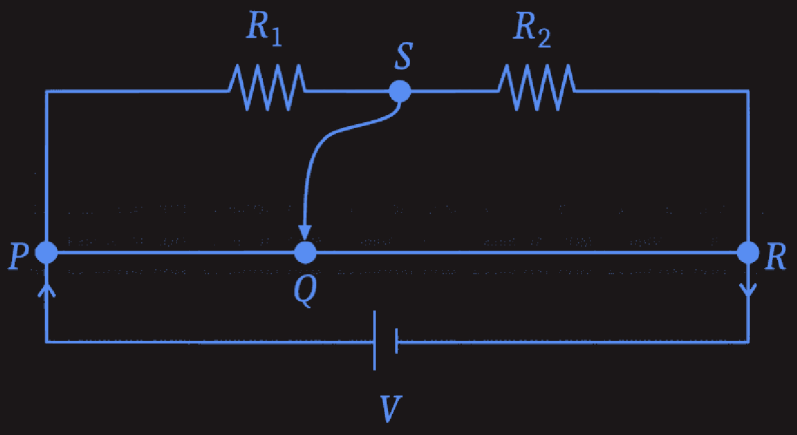


(A) 2 (B) $2\sqrt{5}$ (C) 1 (D) 3

Q70. MCQ (Single Correct Answer)

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A meter bridge with two resistances R_1 and R_2 as shown in figure was balanced (null point) at 40 cm from the point P. The null point changed to 50 cm from the point P, when 16Ω resistance is connected in parallel to R_2. The values of resistances R_1 and R_2 are ____ .



(A) $R_2 = 8\Omega, R_1 = \frac{16}{3}\Omega$

(B) $R_2 = 12\Omega, R_1 = \frac{12}{3}\Omega$

(C) $R_2 = 4\Omega, R_1 = \frac{4}{3}\Omega$

(D) $R_2 = 16\Omega, R_1 = \frac{16}{3}\Omega$

Q71. Numerical

Inductance of a coil with 10^4 turns is 10 mH and it is connected to a dc source of 10 V with internal resistance of 10Ω. The energy density in the inductor when the current reaches $\left(\frac{1}{e}\right)$ of its maximum value is $\alpha\pi \times \frac{1}{e^2}\ \mathrm{J/m^3}$. The value of α is

____ .

$\left(\mu_0 = 4\pi \times 10^{-7}\mathrm{Tm/A}\right)$.

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Q72. Numerical

The electric field of a plane electromagnetic wave, travelling in an unknown nonmagnetic medium is given by,

$E_y = 20\sin\left(3 \times 10^6 x - 4.5 \times 10^{14}\mathrm{t}\right)\mathrm{V/m}$

(where x, t and other values have S.I. units). The dielectric constant of the medium is ____

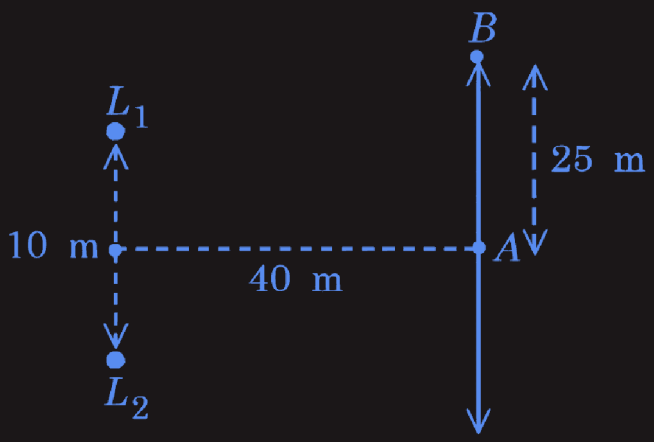
(speed of light in free space is $3 \times 10^8\ \mathrm{m/s}$)

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Q73. Numerical

Two loudspeakers $(L_1$ and $L_2)$ are placed with a separation of 10 m , as shown in figure. Both speakers are fed with an audio input signal of same frequency with constant volume. A voice recorder, initially at point A, at equidistant to both loudspeakers, is moved by 25 m along the line AB while monitoring the audio signal. The measured signal was found to undergo 10 cycles of minima and maxima during the movement. The frequency of the input signal is ____ Hz (Speed of sound in air is $324\ \mathrm{m/s}$ and $\sqrt{5} = 2.23$)

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Q74. Numerical

A parallel beam of light travelling in air (refractive index 1.0) is incident on a convex spherical glass surface of radius of curvature 50 cm . Refractive index of glass is 1.5 . The rays converge to a point at a distance $x\ \mathrm{cm}$ from the centre of the curvature of the spherical surface. The value of x is ____ cm .

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Q75. Numerical

A circular disc has radius R_1 and thickness T_1. Another circular disc made of the same material has radius R_2 and thickness T_2. If the moment of inertia of both discs are same and $\frac{R_1}{R_2} = 2$ then $\frac{T_1}{T_2} = \frac{1}{\alpha}$. The value of α is ____ .

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Answer Key

Chemistry

1. **C**	2. **D**	3. **A**	4. **B**	5. **A**
6. **D**	7. **D**	8. **B**	9. **A**	10. **D**
11. **A**	12. **C**	13. **B**	14. **B**	15. **A**
16. **A**	17. **C**	18. **D**	19. **A**	20. **C**
21. **43**	22. **66**	23. **33**	24. **1303**	25. **4**

Mathematics

26. **C**	27. **D**	28. **C**	29. **D**	30. **D**
31. **D**	32. **B**	33. **B**	34. **D**	35. **D**
36. **A**	37. **A**	38. **D**	39. **D**	40. **D**
41. **B**	42. **A**	43. **C**	44. **B**	45. **A**
46. **4**	47. **18**	48. **660**	49. **49**	50. **16**

Physics

51. **B**	52. **D**	53. **C**	54. **C**	55. **A**
56. **C**	57. **D**	58. **A**	59. **D**	60. **B**
61. **B**	62. **D**	63. **D**	64. **B**	65. **D**
66. **D**	67. **C**	68. **B**	69. **A**	70. **A**
71. **20**	72. **4**	73. **600**	74. **100**	75. **16**

For numerical answers given as a range, any value within the range is correct.

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